PROBLEM 1:

A farmer owns 500 hectares of land in an arid region. The state government gives him up to 1,000,000 cubic metres of water for irrigation each year. In addition, he may purchase up to an additional 300,000 cubic metres of water per annum at a cost of $0.04 per cubic metre.

He grows corn, peas, and onions. The net revenue per hectare of each commodity (excluding the cost of purchased water, if any) and the water requirement in cubic metres per hectare are:

Commodity	Revenue per Hectare	Water Requirement (cubic metres per hectare)
Corn	$200	3500
Peas	$400	6700
Onions	$300	2000

He wishes to diversify his crop in case one commodity suffers an unanticipated fall in price. Therefore, no commodity may occupy more than 50% of the total area planted, nor may any commodity occupy less than 10% of the total area planted.

SOLUTION 1:

We define:

C = the number of hectares of corn planted
P = the number of hectares of peas planted
O = the number of hectares of onions planted
W_1 = the number of cubic metres of free water used
W_2 = the number of cubic metres of purchased water used
A = total area planted (hectares)

	maximize	$200C + 400P + 300O - 0.04W_2$		
	subject to			
Water	Balance	$3500C + 6700P + 2000O - W_1 - W_2$	$=$	0
Water	Free	W_1	$\leq$	1,000,000
Available	Purchased	W_2	$\leq$	300,000
Area	Balance	$C + P + O - A$	$=$	0
"	Amount	A	$\leq$	500
no more	Corn	$C - 0.5A$	$\leq$	0
than	Peas	$P - 0.5A$	$\leq$	0
50%	Onions	$O - 0.5A$	$\leq$	0
no less	Corn	$C - 0.1A$	$\geq$	0
than	Peas	$P - 0.1A$	$\geq$	0
10%	Onions	$O - 0.1A$	$\geq$	0
		all variables must be	$\geq$	0

PROBLEM 2:

Columbian, Peruvian, and Nigerian coffee beans can be purchased for $1.20, $1.00, and $0.90 per kilogram respectively. From these sources a company makes a "regular" and a "premium" blend of coffee, which sell for $1.30 and $1.60 per kilogram respectively. The regular blend contains at least 10% (by mass) Columbian beans, and at least 20% Peruvian beans. The premium blend contains at least 50% Columbian beans, and no more than 15% Nigerian beans. The maximum market demand is for 200,000 kilograms of regular coffee and for 130,000 kilograms of premium coffee.

SOLUTION 2:

We define R and P as respectively the number of kilograms of regular and premium coffee blended, C, V, and N as respectively the number kilograms of Columbian, Peruvian, and Nigerian beans used, and CR, CP, VR, VP, NR, and NP as respectively the number of kilograms of C, V, and N beans used to blend R and P.

	maximize	$1.3R+1.6P-1.2C-V-0.9N$			
	subject to				
Balance	Columbian	$CR+CP-C$	$=$	0	
on	Peruvian	$VR+VP-V$	$=$	0	
Inputs	Nigerian	$NR+NP-N$	$=$	0	
Balance on	Regular	$CR+VR+NR-R$	$=$	0	
Outputs	Premium	$CP+VP+NP-P$	$=$	0	
Columbian	in Regular	$CR-0.1R$	$\geq$	0	
Peruvian	in Regular	$VR-0.2R$	$\geq$	0	
Columbian	in Premium	$CP-0.5P$	$\geq$	0	
Nigerian	in Premium	$NP-0.15P$	$\leq$	0	
Market	Regular	R	$\leq$	$200,000$	
Demand	Premium	P	$\leq$	$130,000$	
		all variables must be	$\geq$	0	

PROBLEM 3:

City bus drivers work two three and a half-hour shifts per day. In some cases, the two shifts are consecutive (effectively one seven-hour shift), but usually they are not. Because of the inconvenience of breaking up the day, those who work non-consecutive shifts are paid a $15 per day bonus. All bus drivers earn a base rate of $25 per hour. The bus company has the following daily requirements:

Time of Day	Minimum Number of Drivers Needed
5:30 a.m. to 8:59 a.m.	150
9:00 a.m. to 12:29 p.m.	80
12:30 p.m. to 3:59 p.m.	90
4:00 p.m. to 7:29 p.m.	160
7:30 p.m. to 10:59 p.m.	70

Subject to meeting all its requirements for drivers, the bus company wishes to minimize its daily labour cost (regular and bonus).

SOLUTION 3:

The question gives both an hourly rate ($25 per hour), and a daily bonus ($15) paid to some of the drivers. The easiest thing to do is to compute a daily rate paid to all drivers ($25 per hour $\times$ 7 hours per day = $175), then both the $175 and the $15 are on a daily basis. We let D_1 = the total number of drivers, and D_2 = the number of drivers who work non-consecutive shifts.

We index the shifts in chronological order from 1 to 5. We define $X_{i,j}$ = the number of drivers who work on shift i, and then shift j, where $j > i$, i.e. $X_{1,2}$, $X_{1,3}$, $X_{1,4}$, $X_{1,5}$, $X_{2,3}$, $X_{2,4}$, $X_{2,5}$, $X_{3,4}$, $X_{3,5}$, and $X_{4,5}$. The shifts which are consecutive are $X_{1,2}$, $X_{2,3}$, $X_{3,4}$, and $X_{4,5}$; the non-consecutive shifts are $X_{1,3}$, $X_{1,4}$, $X_{1,5}$, $X_{2,4}$, $X_{2,5}$, and $X_{3,5}$.

minimize		$175D_1 + 15D_2$		
subject to				
Balance, all drivers		$X_{1,2}+X_{1,3}+X_{1,4}+X_{1,5}$ $+X_{2,3}+X_{2,4}+X_{2,5}+X_{3,4}$ $+X_{3,5}+X_{4,5}-D_1$	$=$	0
Balance, those on non-consecutive shifts		$X_{1,3}+X_{1,4}+X_{1,5}+X_{2,4}$ $+X_{2,5}+X_{3,5}-D_2$	$=$	0
5:30 – 8:59 am	1	$X_{1,2}+X_{1,3}+X_{1,4}+X_{1,5}$	$\geq$	150
9:00 – 12:29 pm	2	$X_{1,2}+X_{2,3}+X_{2,4}+X_{2,5}$	$\geq$	80
12:30 – 3:59 pm	3	$X_{1,3}+X_{2,3}+X_{3,4}+X_{3,5}$	$\geq$	90
4:00 – 7:29 pm	4	$X_{1,4}+X_{2,4}+X_{3,4}+X_{4,5}$	$\geq$	160
7:30 – 10:59 pm	5	$X_{1,5}+X_{2,5}+X_{3,5}+X_{4,5}$	$\geq$	70

all variables $\in \{0, 1, 2, \ldots\}$

Alternatively, the first two constraints could be written using summation notation:

$$\sum_{i=1}^{4} \sum_{j=i+1}^{5} X_{i,j} - D_1 = 0$$

and

$$\sum_{i=1}^{3} \sum_{j=i+2}^{5} X_{i,j} - D_2 = 0$$

The latter equation could also be written as:

$$\sum_{i=1}^{4} X_{i,i+1} + D_2 - D_1 = 0$$

PROBLEM 4:

Wood Products Limited buys fine hardwoods from around the world from which they make specialized products for the quality furniture market. Two of their products are two types of spindles.

A type 1 spindle requires 6 cuts, then 4 minutes of polishing, followed by 6.5 minutes of varnishing. A type 2 spindle requires 15 cuts, then 4 minutes of polishing, followed by 4.75 minutes of painting. There is one cutting machine which can operate up to 135 cuts per hour. There is one polishing machine – allowing for maintenance it can operate up to 54 minutes per hour. Both the varnish and paint shops can only handle one spindle at a time. Because of a periodic need for high volume ventilation, the varnish and paint shops cannot be operated continuously. These shops are available for production 58.5 and 57 minutes per hour, respectively.

For each type 1 spindle produced, the company obtains a contribution to profit of $3. For each type 2 spindle produced, the contribution to profit is $4. How many spindles of each type should be produced each hour so that the total contribution to profit is maximized?

We define:

X_1 — the number of type 1 spindles produced per hour

X_2 — the number of type 2 spindles produced per hour.

$$
\begin{array}{rrcrcrl}
\text{maximize} & 3X_1 & + & 4X_2 & & & \\
\text{subject to} & & & & & & \\
\text{Cutting} & 6X_1 & + & 15X_2 & \leq & 135 & (1) \\
\text{Polishing} & 4X_1 & + & 4X_2 & \leq & 54 & (2) \\
\text{Varnishing} & 6.5X_1 & & & \leq & 58.5 & (3) \\
\text{Painting} & & & 4.75X_2 & \leq & 57 & (4) \\
 & & & & & & \\
 & X_1 & , & X_2 & \geq & 0 &
\end{array}
$$

The following table shows the solution provided by Excel Solver:

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$A$3	OFV	0	46.5

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$5	Spindles/Hour Type 1	0	7.5	Contin
$C$5	Spindles/Hour Type 2	0	6	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$8	Cutting	135	$D$8<=$F$8	Binding	0
$D$9	Polishing	54	$D$9<=$F$9	Binding	0
$D$10	Varnishing	48.75	$D$10<=$F$10	Not Binding	9.75
$D$11	Painting	28.5	$D$11<=$F$11	Not Binding	28.5

The following is the sensitivity report:

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$5	Spindles/Hour Type 1	7.5	0	3	1	1.4
$C$5	Spindles/Hour Type 2	6	0	4	3.5	1

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$8	Cutting	135	0.111111111	135	54	13.5
$D$9	Polishing	54	0.583333333	54	3.6	18
$D$10	Varnishing	48.75	0	58.5	1E+30	9.75
$D$11	Painting	28.5	0	57	1E+30	28.5

1. What happens when the price of type 2 spindles increases by $2 per spindle?
2. Suppose the varnishing shop is available for production for 53.5 minutes (5 minutes less). What would you say about the optimal solution and profit?
3. Suppose the cutting machine can operate up to 162 cuts per hour now. What would you say about the optimal solution and the profit?
4. The demand for type 1 spindle has been going up while there has been less demand for type 2 spindle recently. In order to “manage demand”, Wood Products Limited has decided to change prices. They plan to increase the price of type 1 by $0.4, and decrease the price of type also by $0.4. How do you think this will impact the profit?

SOLUTION 4:

This is less than the “allowable increase of $3.5”, so the optimal solution remains the same, and the profit goes up by $12 (2x6).

5 minutes is less than the “9.75 minutes of allowable decrease”, therefore the shadow price can be used and the profit does not change (as the shadow price is zero for this nonbinding constraint). In this case, one can also conclude that the optimal solution remains the same (as the original solution is still feasible and the objective function value is the same after this change in the right hand side).

The increase of 27 cuts per hour is within the allowable range (allowable increase is 54). The shadow price on the cutting constraint is 0.111, therefore the profit will increase by $3 (0.111x27). We cannot tell how the “optimal solution” would change just by looking at the sensitivity report.

Since you’re simultaneously changing the prices, you need to see if the 100 rule is applicable or not. Allowable increase for type 1 is 1 and allowable decrease for type 2 is also equal to 1. Therefore, the sum of % changes is 0.4+0.4=0.8, therefore the 100 rule applies. The optimal solution is the same, and your profit goes up by $0.4 (increases by 0.4x7,5=3 but decreases by 0.4x6=2.4).